



Vishnu Bharath

Multimodal researcher and student, with an emphasis on entrepreneurship and robotics, using cutting-edge solutions and transformative technology to build a better world.

✉ mr.vishnubharath@gmail.com ☎ [+1 \(508\) 507 5699](tel:+15085075699) 📍 Massachusetts

Education

New England Innovation Academy High School • 4.17 GPA | 1480/1520 PSAT (99th percentile)
2024 - 2028 (Expected)

- FIRST Robotics Competition Team Founder & Captain (founded 2023)
- AI Ethics Advisory Board Co-Chair (creating & implementing AI policies) (founded 2025), pioneering and implementing ethical AI education policies.
- Debate Club Founder & Captain (founded 2021), competed on National Circuit
- Asian Student Association Founding Member & President (founded 2022), fundraised >\$3k, organized visits to MA State house for meetings with the legislature's AAPI caucus

Relevant/Unique Courses:

Entrepreneurship: Foundations | Managing & Growing Ventures | Buying & Selling Ventures

Innovation Studio: Creative Intelligence | Mechanical Marvels | Maker's Revolution

Math: Calculus 1 | Calculus 2

Stanford Online High School High School - Part Time

August 2025 - Present

Stanford University Online High School Program - Accepted Full Time

Multivariable Calculus - A+ (in progress)

Global Online Academy High School - Part Time

September 2023 - May 2025

Elective Courses

Number Theory - A+ | Computer Science II: Java - A | Architecture A+ | Linear Algebra - A | Microeconomics - A |

Macroeconomics - A+ | Religion and Society - A

Experience

MIT Media Lab Cambridge, MA

Student Researcher @ City Science Group June 2024 - Present

- Learned and implemented advanced robotic control methods (RL, LQR, MCP, PID) on (6DOF) bipedal and (16DOF) quadruped last-meter delivery robot concepts. Deployed RL control policies on real-world custom platforms, demoing to 20+ industry customers.
- Developed path-planning and low-level control software in ROS, integrating AprilTag localization, SLAM, and a novel multi-robot docking mechanism for cooperative navigation over complex urban obstacles and energy-optimized routing.
- Worked under Masters and PhD candidates on novel Human Robot Interaction published research regarding legibility and predictability of multi-platform robotic motion.
- Working on societal acceptance of robots in human-shared urban areas using social deference and apology stratagems.
- Invited to stay as 1/1 high school student within the lab on a semipermanent basis (2028+), between City Science @ Taipei and Cambridge.

FIRST Robotics September 2023 - Present

Captain <-- Business, Programming Subteam Lead

- Co-founded (1/5) FRC Team #9644, built fully student-led program in 10 weeks for the 2024 season. Team ranked top 5% globally, qualified for and competed at World Championship 3/3 seasons.
- Led team to 2026 New England Champions in first year as Driver & Captain.
- Led software vision, simulation, and autonomous routine. Utilized advanced control methods (PID, Feedforward, Constraint-bound trapezoidal motion profiling, etc.) along with advanced optimization techniques for optimal robot performance.
- Led fundraising (\$150K+ over 3 years), non-robot awards (5 total), and outreach efforts (connected with >70k people).

NEIA School Store Marlborough, MA

CEO <-- CTO August 2023 - November 2025

- Scaling from zero to over \$25,000 in revenue YoY, achieving full profitability.
- Created the website, product catalog, launch content, presentations, payment processing infrastructure, etc.
- Expanded from a digital store to a physical presence while leading a team of seven.

Publications

Do Robots Need Body Language? Comparing Communication Modalities for Legible Motion Intent in Human-Shared Spaces

March 16th, 2026

Johnathan Cohen, Kye Shimizu, Vishnu Bharath, Yu-Lun Song, Kent Larson, Pattie Maes

Published in ACM and IEEE for HRI '26.

We propose developing and evaluating a set of expressive motor behaviors for legged robots (e.g., quadrupeds) that communicate intent through motion and posture in urban, human-shared spaces. (MIT Media Lab)

<https://doi.org/10.1145/3776734.3794408>

Unique Projects

Pristiq Summer 2023 **Gear Grease** April 2024

Created proposal for new, blockchain-based, freelance website: Pristiq. Implemented blockchain for transparent and low-cost payment systems, with other task-oriented implementations. Proposal included advertising strategy, finance & budgeting plans, potential advertisement, backend design, frontend simulation with Figma.

Led a team of 4 to develop a project for the MIT Day of AI, creating a proposal for a sustainability oriented video game: Gear Grease. Implemented unique idea of Natural Language Processing and Large Language Model API integration to create NPC dialogue.

Presented in front of Dean Amir Reza, Prof. Vincent Onyemah, and Prof. Kristen Getchell (Babson). Awarded 1st place in Rocket Pitch Competition.

Invited to speak as a guest presenter at the Day of AI event, hosted at the Boston Museum of Science, alongside MIT RAISE director Cynthia Breazeal, MIT Media Lab director Dava Newman, and other distinguished leaders. >8,000 teachers (teaching classes) with virtual program, with participants from 110+ countries.

Certifications & Proficiencies

Fundamentals of Deep Learning NVIDIA Practical course on model architecture, training, optimization, and deployment using modern GPU-accelerated workflows.	Programming Languages: Java, Python, C++ Libraries: Pandas, TF, PyTorch, Numpy, Matplotlib Tools: Git, Docker, AWS/Azure	Entrepreneurial Certificate Babson College Participated in lectures & workshops hosted by 10+ Babson full-time professors across a 2 week period.	Bootcamp: Entrepreneurship 101 MIT MITx (edX) course on Customer & Market analysis. Score: 100/100 4-Day Intensive with Senior Lecturer Erdin Beshimov
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Interests

Fencing Saber	Volunteering Meher Foundation	Motorsports Go-Karts, Sim-Racing	Politics & Current Events Canvassing, Phone Banking
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Languages

Mandarin Chinese	Hindi	Telegu	English
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